

Switching and VLANs

Q1. Network diagram with VLANs

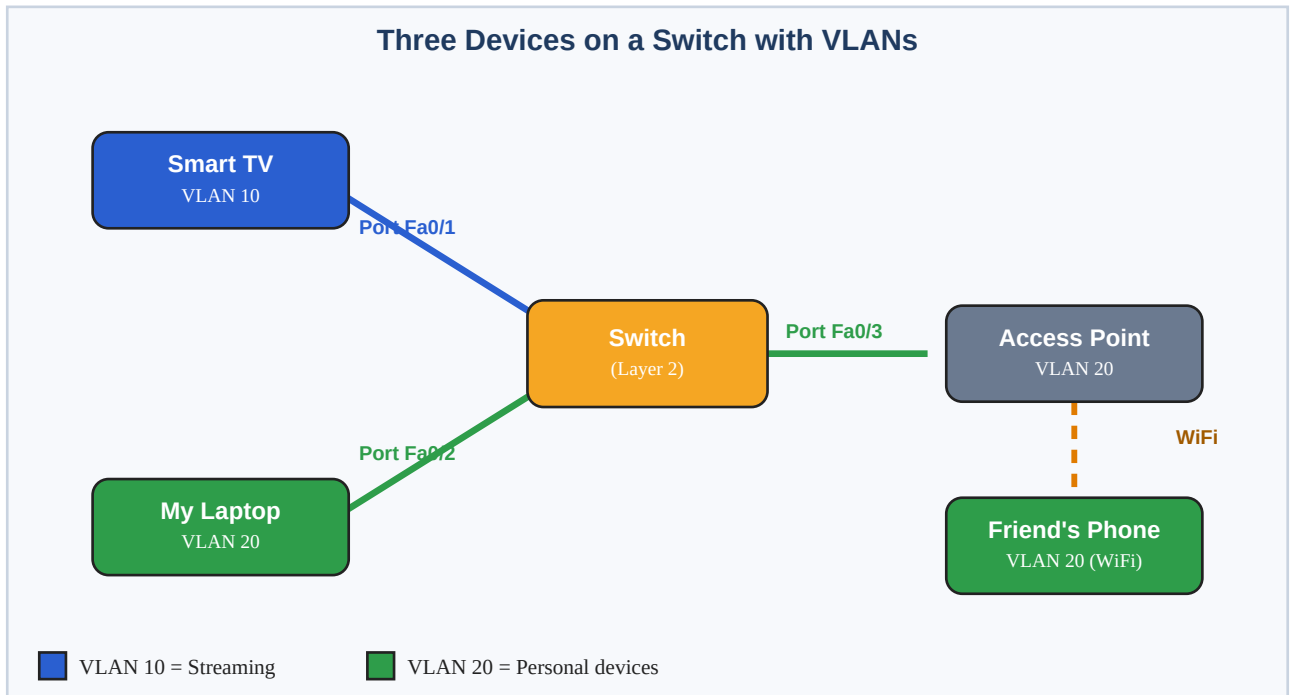


Figure 1: Smart TV, laptop and friend's phone connected to one switch, split into VLANs

Device	VLAN	How it connects
Smart TV	VLAN 10 (Streaming)	Ethernet cable to switch port Fa0/1
My laptop	VLAN 20 (Personal)	Ethernet cable to switch port Fa0/2
Friend's phone	VLAN 20 (Personal)	WiFi to an access point, which is on switch port Fa0/3

Effect: the laptop and phone can talk to each other because they are in the same VLAN. The smart TV is separated from them, so a device in the TV's VLAN cannot directly reach the laptop or phone. Traffic between VLANs needs a router.

Q2. Two VLANs on a simulated switch (Cisco Packet Tracer)

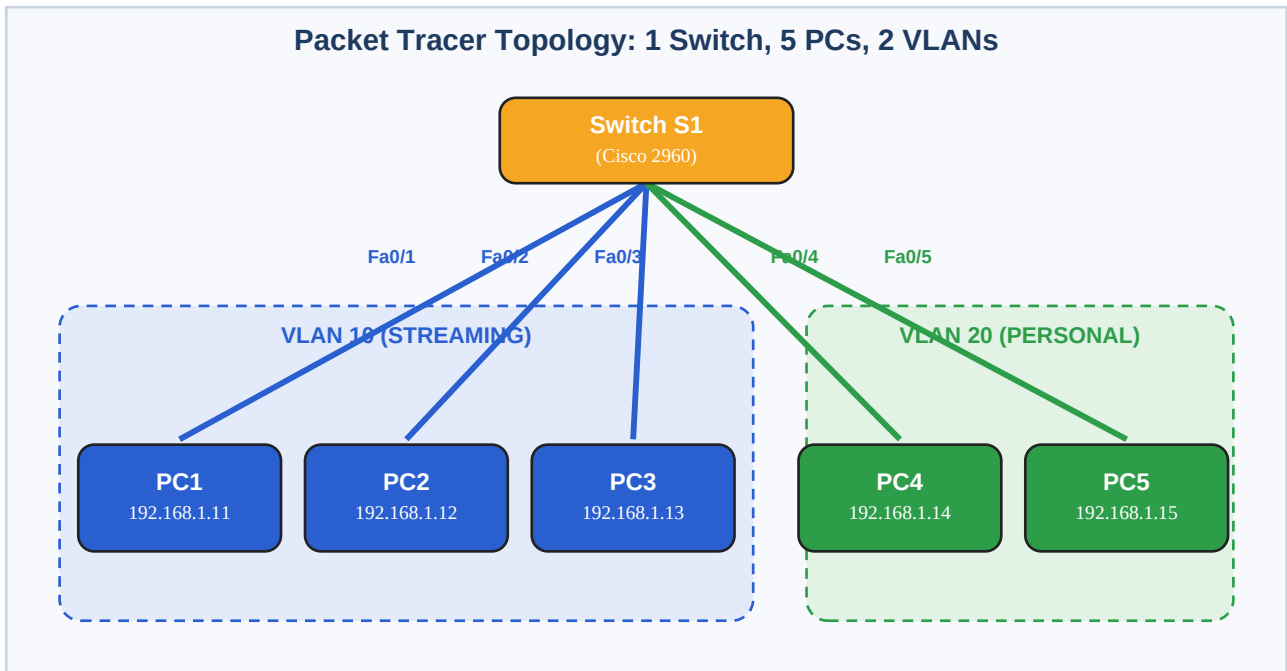


Figure 2: Topology to build in Packet Tracer. Use a 2960 switch, 5 PCs and Copper Straight-Through cables.

Steps

Step	Action
1	Add one Cisco 2960 switch and five PCs, then connect PC1 to Fa0/1, PC2 to Fa0/2, PC3 to Fa0/3, PC4 to Fa0/4 and PC5 to Fa0/5.
2	On each PC: Desktop, then IP Configuration, and enter the IP address and mask 255.255.255.0 from the figure. No gateway is needed.
3	Open the switch CLI and type the commands below.

Switch configuration

```

Switch> enable
Switch# configure terminal
Switch(config)# hostname S1
S1(config)# vlan 10
S1(config-vlan)# name STREAMING
S1(config-vlan)# exit
S1(config)# vlan 20
S1(config-vlan)# name PERSONAL
S1(config-vlan)# exit
S1(config)# interface range fastEthernet 0/1 - 3
S1(config-if-range)# switchport mode access
S1(config-if-range)# switchport access vlan 10
S1(config-if-range)# exit
S1(config)# interface range fastEthernet 0/4 - 5
S1(config-if-range)# switchport mode access
S1(config-if-range)# switchport access vlan 20
S1(config-if-range)# end
S1# show vlan brief
S1# write memory
  
```

Example output of show vlan brief (yours may list more ports)

VLAN	Name	Status	Ports
1	default	active	Fa0/6, Fa0/7, Fa0/8, ... Gig0/1, Gig0/2
10	STREAMING	active	Fa0/1, Fa0/2, Fa0/3
20	PERSONAL	active	Fa0/4, Fa0/5

Connectivity test

Ping from	Ping to	Expected result
PC1 (VLAN 10)	PC2 or PC3 (VLAN 10)	Success: replies received
PC4 (VLAN 20)	PC5 (VLAN 20)	Success: replies received
PC1 (VLAN 10)	PC4 or PC5 (VLAN 20)	Fails: Request timed out (100% loss)
PC4 (VLAN 20)	PC1, PC2 or PC3 (VLAN 10)	Fails: Request timed out (100% loss)

All PCs use the same subnet (192.168.1.0/24), so the ping fails only because the switch keeps the two VLANs separate. On a PC, use Command Prompt: `ping 192.168.1.14`. Take screenshots of the topology, show vlan brief and the ping results.

Q3. Moving PC3 from VLAN 10 to VLAN 20

Using the same topology, I move **PC3** (port Fa0/3, IP 192.168.1.13) from VLAN 10 to VLAN 20.

Before

```
S1# show vlan brief
10  STREAMING    active    Fa0/1, Fa0/2, Fa0/3
20  PERSONAL     active    Fa0/4, Fa0/5

PC3> ping 192.168.1.11    -> Success (PC1 is in VLAN 10)
PC3> ping 192.168.1.14    -> Request timed out (PC4 is in VLAN 20)
```

Change

```
S1# configure terminal
S1(config)# interface fastEthernet 0/3
S1(config-if)# switchport access vlan 20
S1(config-if)# end
S1# write memory
```

After

```
S1# show vlan brief
10  STREAMING    active    Fa0/1, Fa0/2
20  PERSONAL     active    Fa0/3, Fa0/4, Fa0/5

PC3> ping 192.168.1.14    -> Success (PC4 is now in the same VLAN 20)
PC3> ping 192.168.1.11    -> Request timed out (PC1 is in VLAN 10)
```

Conclusion: only the port command was changed, and PC3's IP address stayed the same, yet its reachability changed. This shows that the VLAN, not the IP address, decides which devices PC3 can reach.

Q4. Assign 10 ports to VLAN 30 with one range command (Cisco Packet Tracer)

The **interface range** command applies the same settings to ports Fa0/1 to Fa0/10 together, so no command is repeated 10 times.

```
Switch> enable
Switch# configure terminal
Switch(config)# vlan 30
Switch(config-vlan)# name LAB
Switch(config-vlan)# exit
Switch(config)# interface range fastEthernet 0/1 - 10
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 30
Switch(config-if-range)# end
Switch# show vlan id 30
Switch# write memory
```

Optional: generating the commands with a loop (Python) and pasting them into the CLI

```
commands = ["vlan 30", " name LAB"]
for port in range(1, 11):          # ports Fa0/1 to Fa0/10
    commands += [f"interface fastEthernet 0/{port}",
                  " switchport mode access",
                  " switchport access vlan 30"]
print("\n".join(commands))
```

The range command is the preferred method because it is shorter and works directly on the switch. The loop is useful when many switches or irregular port lists must be configured.